

Lecture 4, Hyperbolic groups

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Definition: A metric space X is *proper* if closed balls of finite radius in X are compact. The action of a group Γ on a metric space X is *cocompact* if X/Γ is compact in the quotient topology.

Definition: If $A, B \subseteq X$ are compact, then $d_{Haus}(A, B) = \max\{\sup_{a \in A} \inf_{b \in B} d(a, b), \sup_{b \in B} \inf_{a \in A} d(a, b)\}$ is the *Hausdorff distance* between A and B .

The Svarc-Milnor Lemma: Let X be a proper geodesic metric space. Let Γ act cocompactly and properly discontinuously on X . (Properly discontinuous means that $\forall$ compact $K \subseteq X$, $|\{\gamma \in \Gamma \mid \gamma K \cap K \neq \emptyset\}| < \infty$). Then Γ is finitely generated and, for any $x_0 \in X$, the map $\Gamma \rightarrow X$ defined by $\gamma \mapsto \gamma x_0$ is a quasi-isometry (where Γ is equipped with the word metric).

Proof: We may assume that Γ is infinite and X is non-compact. Let R be large enough that Γ -translates of $B = B(X, R)$ cover X . Set $S = \{s \in \Gamma \setminus \{1\} \mid s\bar{B} \cap \bar{B} \neq \emptyset\}$. Let $r = \inf\{d(\bar{B}, \gamma\bar{B}) \mid \gamma \in \Gamma, \gamma \notin S \cup \{1\}\}$. Let $\lambda = \max_{s \in S} d(x_0, sx_0)$.

We want to prove that:

- (a) S generates Γ ,
- (b) $\forall \gamma \in \Gamma, \underbrace{\lambda^{-1}d(x_0, \gamma x_0)}_{\text{i}} \leq l_S(\gamma) \leq \frac{1}{r} \underbrace{d(x_0, \gamma x_0) + 1}_{\text{ii}},$
- (c) $\forall x \in X, \exists \gamma \in \Gamma$ such that $d(x, \gamma x_0) \leq R$.

Note: $d_S(1, \gamma) = l_S(\gamma)$ and $d_S(\gamma, \delta) = l_S(\gamma^{-1}\delta)$.

(c) is obvious.

(b-i) is also obvious.

To complete the proof we need to show (a) and (b – ii).

Assume $\gamma \notin S \cup \{1\}$. Let k be such that

$R + (k - 1)r \leq d(x_0, \gamma x_0) < R + kr$. As $\gamma \notin S \cup \{1\}$, $k > 1$.

Choose $x_1, \dots, x_{k+1} = \gamma x_0$, such that $d(x_0, x_1) < R$ and

$d(x_i, x_{i+1}) < r$ for each $i > 0$. Choose $1 = \gamma_0, \gamma_1, \dots, \gamma_{k-1}, \gamma_k = \gamma$ such that $x_{i+1} \in \gamma_i \bar{B}$ for each i . Let $s_i = \gamma_{i-1}^{-1} \gamma_i$, so $\gamma = s_1 \cdots s_k$.

Now $d(\bar{B}, s_i \bar{B}) \leq d(\gamma_{i-1}^{-1} x_i, s_i \gamma_{i-1}^{-1} x_{i+1}) = d(x_i, x_{i+1}) < r$, so $s_i \in S \cup \{1\}$. Therefore S generates Γ .

Also, $l_S(\gamma) \leq k \leq \frac{1}{r} d(x_0, \gamma x_0) + \frac{r-R}{r} < \frac{1}{r} d(x_0, \gamma x_0) + 1$ as required.

Corollary: If $K \leq \Gamma$ is a finite index subgroup of finitely generated group then K is quasi-isometric to Γ .

Quasi-geodesics

In the following, X is always a geodesic metric space.

Recall: A quasi-geodesic is a quasi-isometrically embedded interval.

Theorem

For all $\delta \geq 0$, $\lambda \geq 1$, $\epsilon \geq 0$, there exists $R = R(\delta, \lambda, \epsilon)$ with the following property: if X is a δ -hyperbolic metric space, $c : [a, b] \rightarrow X$ is a (λ, ϵ) -quasi-geodesic, and $[c(a), c(b)]$ is any geodesic from $c(a)$ to $c(b)$, then $d_{\text{Haus}}(\text{im } c, [c(a), c(b)]) \leq R(\delta, \lambda, \epsilon)$.

Quasi-geodesics

Corollary

A geodesic metric space X is hyperbolic if and only if for every $\lambda \geq 1$, $\epsilon \geq 0$, there exists an M such that every (λ, ϵ) -quasi-geodesic triangle is M -slim.

Corollary

If X is a δ -hyperbolic space, Y is quasi-geodesic space, and

$$f : Y \rightarrow X$$

be a (k, c) -quasi-isometry, then Y is $k(2R + \delta) + c$ -hyperbolic.

Corollary

Hyperbolicity is a quasi-isometry invariant of geodesic metric spaces.

Quasi-geodesics

Proposition: Equivalent definition of a hyperbolic group: A group Γ is hyperbolic if it acts properly discontinuously and cocompactly by isometries on a proper hyperbolic metric space.

Examples:

- a) Free groups;
- b) $\mathbb{Z}^n$ is not hyperbolic for $n > 1$;
- c) Let M be any closed hyperbolic manifold. Then $\pi_1(M)$ is word-hyperbolic.

Quasiconvex subgroups

Definition A subspace Y of a geodesic metric space X is *quasiconvex* if there exists a $K \geq 0$ such that, for all $y_1, y_2 \in Y$ and for all $x \in [y_1, y_2]$, $d(x, Y) \leq K$.

Definition: A subgroup H of a hyperbolic group is called *quasiconvex* if it is a quasiconvex space of some (any) Cayley graph of Γ .

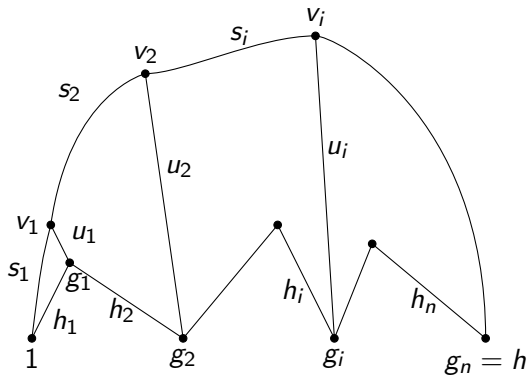
Example: Consider $\mathbb{Z}^2$ with the l^1 -metric. Then the diagonal subgroup $\Delta \subset \mathbb{Z}^2$ is not quasiconvex (though it is quasi-isometrically embedded).

Quasiconvex subgroups

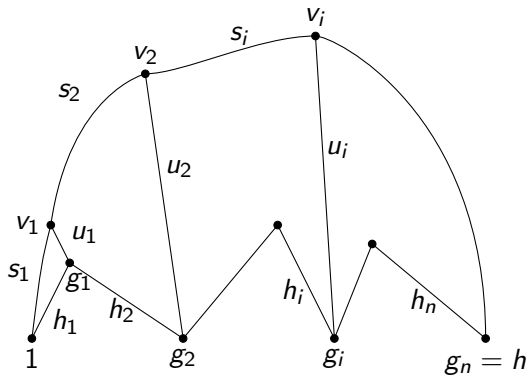
Corollary: Suppose that Γ is a word-hyperbolic group and H is a subgroup. Then H is quasiconvex in some (any) Cayley graph of Γ if and only if H is finitely generated and $H \hookrightarrow \Gamma$ is a quasi-isometric embedding.

Quasiconvex subgroups

Proof: ($\Leftarrow$) is immediate from the Theorem. For the other direction, fix a generating set S for Γ , assume H is quasiconvex in the Cayley graph of Γ with constant K , and let h be in H . Consider a geodesic in the Cayley graph of Γ from 1 to h , which we can take to be of the form $s_1 \dots s_n$ for s_i in S . Let v_i be the vertices of this geodesic, so $v_i = s_1 \dots s_i$.



Quasiconvex subgroups



$\forall v_i$ there exists g_i in

H s. t. $d(v_i, g_i) \leq K$. Take $g_0 = 1$ and $g_n = h$. Let $h_i = g_{i-1}^{-1} g_i$, so $h = h_1 \dots h_n$. Let $u_i = v_i g_i^{-1}$, $l_S(u_i) \leq K$. For each i , we have that $h_i = u_i s_i u_{i+1}^{-1}$ and so $l_S(h_i) \leq 2K + 1$. Therefore, H is generated by $T = B(1, 2K + 1) \cap H$, a finite set. Furthermore, $l_T(h) \leq n = l_S(h)$.

$l_S(h) \leq (2K + 1)l_T(h)$ (as each element of T has S -length at most $2K + 1$) so $H \leq_{q.i.} \Gamma$.

Quasiconvex subgroups

Definition: Let $H \leq G$. A homomorphism $\phi : GH$ is a retraction if $\phi(h) = h$ for $h \in H$. H is a retract of G .

Exercise: If $G \xrightarrow{\phi} H$ is a retraction and G is finitely generated, then the inclusion $H \hookrightarrow G$ is a quasi-isometric embedding. (Hint: you can choose a generating set S for G such that $\phi(s) = s$ or $\phi(s) = 1$ for all $s \in S$.)

Example: Marshall Hall's Theorem implies that every finitely generated subgroup of a free group is a retract of a finite-index subgroup, so every finitely generated subgroup of a free group is quasiconvex.

Theorem

- 1) *Quasiconvex subgroups of hyperbolic groups are hyperbolic.*
- 2) *Intersection of two quasi-convex subgroups is quasi-convex.*
- 3) *Membership problem in a quasiconvex subgroup is decidable.*

Quasiconvex subgroups

Example

In a hyperbolic group, if $g \in G$ is infinite order, then the centralizer of g is quasiconvex.

Example

No hyperbolic group contains $\mathbb{Z}^2$ as a subgroup. In fact, it cannot contain a Baumslag-Solitar subgroup,

$$\langle a, b \mid b^{-1}a^mb = a^n \rangle.$$

Quasiconvex subgroups

Definition

Let $G = \langle S | R \rangle$. We say that the symmetric presentation is a Dehn presentation if for any reduced word w with $w = 1$ in G , there exists a relator $r \in R$ so that $r = r_1 r_2$,

$$l(r_1) > l(r_2),$$

and

$$w = w_1 r_1 w_2.$$

In other words, any word that represents the identity in G contains more than one half of a relator, and so it can be shortened. A word which cannot be further reduced or shortened by this method (replacing r_1 by r_2^{-1} , a shorter word) is called Dehn reduced.

Quasiconvex subgroups

If G has a finite Dehn presentation (G is finitely generated, R is finite) then you can check all subwords of length at most

$$N = \max \{l(r) \mid r \in R\},$$

to see if a reduction can be made. This procedure for solving the word problem is called Dehn's Algorithm, originally created by Max Dehn in 1910 to solve the word problem in surface groups. Its run time is $O(|w|^2)$ in its simplest iteration. There are at most $|w| - N$ subwords in each step, and at most $|w|$ steps in the reduction.

Quasiconvex subgroups

Example

Let $G = \langle a, b, c, d \mid [a, b] [c, d] \rangle$. Let R be the symmetrized set of generators, so that R contains all cyclic conjugates of $[a, b] [c, d]$ and its inverse.

What about other hyperbolic groups?

Definition

A path γ in a metric space X is called a k -local geodesic if every subpath of length k is a geodesic.

Example

On a sphere of radius 1, a great circle is a π -local geodesic.

Quasiconvex subgroups

Lemma

Let G be δ -hyperbolic group and let γ be a 4δ -local geodesic. Let g be the geodesic between the endpoints of γ (called γ_+ and γ_-). Assume

$$l(g) > 2\delta,$$

and let r and s be points on γ and g respectively, both distance 2δ from γ_+ . Then

$$d(r, s) \leq \delta.$$

The proof here is by induction on the length of γ and uses the thin triangle property multiple times.

Quasiconvex subgroups

Theorem

If γ is a 4δ –local geodesic in a δ –hyperbolic group G , then γ is contained in the 3δ neighborhood of the geodesic between its endpoints.

Quasiconvex subgroups

We will use this theorem to create a shortening algorithm in our hyperbolic group G .

Theorem

Let G be a δ -hyperbolic group with generators S . Let R be equal to the set of words

$$R = \{w : w = 1 \in G, |w| < 8\delta \text{ and } w \text{ represents the identity element in } G\}$$

We aim to show that $\langle S | R \rangle$ is a Dehn presentation for G . In particular G is finitely presented and has a solvable word problem.

Quasiconvex subgroups

Proof.

Take a word w in the generators so that $w = 1$ in G . If the loop w in the Cayley graph is already a 4δ -local geodesic, then it is in the 3δ neighborhood of the origin, so it is already an element of R and we can see that it represents the identity. Now say that w is not a 4δ which is not a geodesic between its endpoints. Replace it in w by the geodesic between its endpoints, call it w_2 . Then the path $w_1 w_2^{-1}$ has length less than 8δ , so that word is in R . Further, we note that w contained w_1 , the longer part of the relator. Now, we can continue this process until we reduced w to a 4δ -local geodesic, a case we have already covered. Thus

$$G = \langle S | R \rangle,$$

is a Dehn presentation. □

Conjugacy problem

Theorem

If Γ is a δ -hyperbolic group and $u = \gamma v \gamma^{-1}$ in Γ , then there exists such an element γ with $\ell(\gamma) \leq M(\ell(u) + \ell(v))$, where M depends only on Γ . Therefore the conjugacy problem in Γ is decidable.

Isomorphism problem

Decidable! (Dahmani, Guirardel)

Stallings-like graph

Theorem: (Khar., Miasnikov, Weil) There is an algorithm to construct a Stallings-like graph for a quasiconvex subgroup H of a hyperbolic group G (if we know that G is hyperbolic).

Previously: Kapovich, Arzhantseva-Olshanski.

Rips construction

There are f.g. non-quasiconvex subgroups of hyperbolic groups such that the membership problem is undecidable. (Rips construction:

Theorem Given any finitely presented group Q and any real number $\lambda > 0$, there is a $C'(\lambda)$ group G and a short exact sequence

$$1 \rightarrow N \rightarrow G \xrightarrow{p} Q \rightarrow 1$$

with N finitely generated.

Theorem (Rips) There is a small cancellation group G such that: G has finitely generated subgroups whose intersection is not finitely generated.

G has a finitely generated but not finitely presented subgroup

The subgroup membership problem in G is not solvable. Indeed, there is a subgroup (N) for which there is no algorithm deciding whether a word in the generating set for G represents an element of the subgroup.

Open questions

1. (Gromov) Given a hyperbolic group G with one topological end (i.e. a freely indecomposable hyperbolic group), does it contain a surface subgroup?
2. Are hyperbolic groups residually finite?
3. Does every hyperbolic group have a proper subgroup of finite index?
4. (A.Miasnikov) Given a finite presentation of a hyperbolic group (which is not necessarily a Dehn presentation), is it possible to find a Dehn presentation for this group in polynomial time?